

AE646 — Individual Contribution Statement & AI Tool Use Declaration

Team PINNacles

Aryamann Srivastava

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Individual Contribution Statement

Team member: Aryamann Srivastava

- **Problem formulation and literature review:** Identified the 2D Darcy flow benchmark in PDEBench as the project focus; reviewed the FNO paper [1] and PDEBench paper [2].
- **Data pipeline:** Implemented `src/download_data.py` and `src/preprocess.py` for downloading and preprocessing PDEBench data; designed the 900/100/200 train/val/test split (a 1000-sample train/val pool plus 200 held-out test samples, drawn with a fixed seed from PDEBench's 10000-sample archive).
- **Model implementation:** Implemented `MLPBaseline`, `SpectralConv2d`, and `FNO2d` in `src/models.py`; verified correctness against the reference FNO codebase.
- **Training:** Implemented `src/train.py` with AdamW and a cosine learning-rate schedule, retaining the checkpoint with the lowest validation relative- L_2 error; ran all three training runs (FNO original, MLP, FNO improved) on the lab GPU VM.
- **Evaluation:** Implemented `src/evaluate.py`, `src/superres_eval.py`, and `src/benchmark_speed.py`; collected all quantitative results.
- **Component profiling:** Designed and implemented `src/benchmark_components.py` with correct GPU synchronisation to measure per-layer execution time.
- **Analysis and ablations:** Implemented `src/eda.py` (dataset analysis and the error-vs-geometry study), `src/ablation_mlp.py` (depth/optimiser) and `src/ablation_preprocess.py` (preprocessing, 3 seeds); ran them on the lab GPU workstation.
- **Stage 3 optimisation study:** Designed the Stage 3 experiment matrix and implemented `src/stage3_train.py` (fixed-budget training with recipe options, D4 augmentation and test-time symmetrisation, multi-resolution evaluation), `src/run_jobs.py` (job queue), `src/data_cache.py`, the vectorised finite-volume solver `src/solvers.py` with its calibration and timing (`src/solver_vs_fno.py`), `src/stage3_latency.py`, `src/stage3_diagnostics.py`, and `src/stage3_aggregate.py/stage3_figures.py`; ran the 142 runs (recipe ablation, architecture sweeps, data scaling, resolution study, five-seed final models) on the lab GPU workstation.
- **Figure generation:** Implemented `src/generate_figures.py`; produced the report figures from stored JSON results.

- **Reports and presentations:** Wrote Stage 1 proposal (`stage1/PINNacles_Stage1_Proposal.tex`), Stage 1 presentation (10 slides), Stage 2 interim report, presentation and code package (`stage2/`, including a single notebook that combines the pipeline scripts), the Stage 3 final presentation (`stage3/PINNacles_Stage3_FinalPresentation`), and this final report (`stage3/PINNacles_Stage3_FinalReport.tex`).

Team PINNacles was formed with four members at the proposal stage (Aryamann Srivastava, Varun Sathaye, Atishay Jain, Vedant Shekhar Tiwari). The implementation, experiments, figures, and reports contained in this repository were carried out by Aryamann Srivastava, whose individual contributions are listed above.

AI Tool Use Declaration

In accordance with the AE646 course policy on generative AI tools, I declare the following use of AI assistance during this project:

1. **Coding assistance and debugging:** Claude Code (Anthropic) assisted with writing and debugging project code, including the data, training and evaluation scripts, the analysis and ablation scripts, and the notebook that combines them. One example is the GPU timing fix in the component profiling script: `time.perf_counter()` measured dispatch time rather than execution time for asynchronous GPU operations, and the fix (a device-synchronisation call before every timer read — `torch.cuda.synchronize` on CUDA, `torch.mps.synchronize` on Apple Silicon) was suggested by the AI assistant. All code was reviewed and accepted by me before being committed.
2. **Running experiments and checking results:** At my direction, Claude Code executed the scripts on the lab GPU workstation (training, ablations, the notebook run) and cross-checked every number quoted in the reports, slides and README against the stored results files and training configs. This surfaced several documentation errors that were then corrected (for example the training-loss and learning-rate-schedule descriptions, the dataset split, and rounded percentage figures).
3. **Report and slide drafting:** Claude Code assisted with LaTeX structure and table formatting, with drafting and editing report text, and with Python (`python-pptx`) and Beamer code to build the Stage 1, Stage 2 and Stage 3 presentations. The technical content, numbers and interpretation were reviewed and verified by me.
4. **Stage 3 experiments:** Claude Code also proposed and implemented the Stage 3 experiment design and scripts listed above, launched the runs on the lab workstation at my direction, and identified that the Stage 2 training recipe (MSE loss, a learning-rate schedule that restarted every 200 steps) and the Stage 2 solver-timing reference were sub-optimal; the Stage 2 speed-up claim was withdrawn in the final report as a result. I reviewed the design, the results and the conclusions.
5. **No AI-generated results:** All numerical results in this report come from real experiments run on real data and hardware. No results were generated or invented by AI tools.

Signature: Aryamann Srivastava

Date: November 2026

References

- [1] Z. Li et al., “Fourier Neural Operator for Parametric Partial Differential Equations,” *ICLR*, 2021.
- [2] M. Takamoto et al., “PDEBench,” *NeurIPS*, 2022.